

ANALYSTLAB AFRICA

Experience Lab Internship Programme

WEEK 1 SUBMISSION

FinTrust Digital Bank — Customer & Transaction Intelligence

Data Analytics Track

Project	FinTrust Financial Intelligence & Digital Banking Support Solution
Week	Week 1 — Understand & Plan
Track	Data Analytics
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Submission	Week 1 Individual Submission

Data note

All FinTrust customer, transaction and financial information used in this project is synthetic and intended for educational/training purposes.

1. Executive Summary

FinTrust Digital Bank is a fictional digital banking organisation seeking to use data, machine learning and generative AI to improve its understanding of customer and transaction behaviour, support risk-related decision-making, improve operational processes and strengthen digital customer support.

As the Data Analytics track, the Week 1 objective is to establish a reliable analytical foundation before dashboard development. This submission documents the business problem, available resources, data structure and quality observations, analytical questions, KPI definitions, proposed dashboard structure, initial objectives, success criteria, risks, dependencies and a Week 2–4 delivery plan.

Week 1 approach

Understand the business problem → profile the available data → define answerable analytical questions → select measurable KPIs → design the dashboard concept → establish the analysis plan.

2. Business Understanding

2.1 Business Problem

FinTrust needs to transform customer and transaction data into useful business intelligence that can help management understand customer behaviour, monitor transaction activity and performance, and identify patterns associated with transactions requiring risk review.

2.2 Why the Problem Matters

- Customer and transaction data can reveal patterns that are difficult to identify from individual records.
- Transaction performance information can help management monitor successful, failed, pending and reversed transactions.
- Channel-level analysis can support decisions about digital banking service performance and customer experience.
- Risk-related analysis can highlight patterns that deserve further review without treating the synthetic risk label as proof of fraud.
- A consistent KPI framework can give stakeholders a common view of customer and transaction performance.

2.3 Business Questions for Management

1. Who are FinTrust's customers and how does transaction behaviour vary across customer segments?
2. How frequently are customers using FinTrust's services, and which transaction types and channels are most active?
3. Which transaction types and channels contribute the greatest transaction value?
4. How do transaction outcomes vary across channels?
5. What customer and transaction characteristics are associated with the synthetic Risk_Review_Flag?
6. Which operational or customer patterns require further investigation?

2.4 Decisions the Analysis Could Support

- Prioritisation of operational improvements based on transaction performance.
- Monitoring and improvement of digital banking channels.

- Customer engagement and service planning using observed customer behaviour.
- Further investigation of transaction patterns associated with the synthetic risk-review label.
- Ongoing performance reporting through a reusable KPI dashboard.

2.5 Stakeholders

Stakeholder	Potential Use of Analytical Outputs
Management / Executives	Overall customer, transaction and KPI visibility.
Operations Team	Transaction success, failure, pending and reversal monitoring.
Customer Experience Team	Customer behaviour, engagement and channel usage.
Digital Banking Team	Channel performance and digital adoption patterns.
Risk Team	Patterns associated with transactions flagged for review.
Data / Analytics Team	Data preparation, analysis, reporting and future modelling.

3. Resource Review

The Week 1 work is based on the approved FinTrust resources and the Week 1 assignment requirements. The main data resources available for this analysis are the Customer dataset, Transaction dataset and Data Dictionary.

Resource	What It Contains	How It Will Be Used	Relevant Limitation / Note
FinTrust Customer Data.xlsx	1,500 customer records and 12 customer fields.	Customer profiling, segmentation and joining customer attributes to transactions.	The dataset is synthetic; customer names are not suitable as unique identifiers.
FinTrust Transaction Data.xlsx	12,000 transaction records and 11 transaction fields.	Transaction activity, value, channel, status and risk-pattern analysis.	Device_Type and Location contain missing values; risk flag is synthetic.
FinTrust Data Dictionary.xlsx	Definitions and business/modelling use for the supplied fields.	Interpret field meaning, data types, joins and appropriate analytical use.	Definitions should guide later cleaning and modelling decisions.
Week 1 Assignment / Project Brief	Business, track and submission requirements.	Guide scope, deliverables, success criteria and weekly planning.	Week 1 explicitly states that extensive cleaning is not yet required.

The Financial Knowledge Base is listed in the programme resources and can be incorporated later for financial terminology and business context when the file is available to the project workspace.

4. Data Understanding & Profiling

4.1 Dataset-Level Profile

Dataset	Records	Columns	Unique Key	Duplicate Rows	Missing Cells	Join Key
Customer	1,500	12	Customer_ID (1,500)	0	0	Customer_ID
Transaction	12,000	11	Transaction_ID (12,000)	0	192	Customer_ID

4.2 Customer Dataset Fields

Field	Data Type	Variable Type	Initial Profile
Customer_ID	Text	Identifier	1,500 unique; no missing
Customer_Name	Text	Name / descriptive	180 unique; no missing
Age	Integer	Numerical	18–65; no missing
Gender	Categorical	Categorical	3 categories; no missing
City	Categorical	Categorical	8 categories; no missing
Customer_Segment	Categorical	Categorical	4 categories; no missing
Account_Type	Categorical	Categorical	3 categories; no missing
Tenure_Months	Integer	Numerical	1–96; no missing
Digital_Engagement_Score	Decimal	Numerical	16.7–100.0; no missing
Monthly_Income_Band	Categorical	Categorical	5 categories; no missing
Preferred_Channel	Categorical	Categorical	3 categories; no missing
Account_Status	Categorical	Categorical	3 categories; no missing

4.3 Transaction Dataset Fields

Field	Data Type	Variable Type	Initial Profile
Transaction_ID	Text	Identifier	12,000 unique; no missing
Customer_ID	Text	Foreign key	1,500 unique; no missing
Transaction_DateTime	Date/Time	Date/time	1 Jan 2026 – 31 Mar 2026
Transaction_Type	Categorical	Categorical	6 categories; no missing
Amount_NGN	Decimal	Numerical	₦100.17–₦693,454.45; no missing
Channel	Categorical	Categorical	5 categories; no missing
Device_Type	Categorical	Categorical	5 observed categories; 96 missing
Location	Categorical	Categorical	8 observed categories; 96 missing
International_Transaction	Categorical	Binary	Yes/No; no missing
Transaction_Status	Categorical	Categorical	4 categories; no missing
Risk_Review_Flag	Categorical	Binary label	Yes/No; no missing

4.4 Key Data Relationships

Customer_ID provides the relationship between the two datasets. The Customer dataset contains 1,500 unique customers, while the Transaction dataset contains 1,500 distinct Customer_ID values. All transaction Customer_ID values were found in the Customer dataset, with no orphan transaction IDs identified.

Each customer has at least one transaction. There are 12,000 transactions across 1,500 customers, giving an overall average of 8.0 transactions per customer.

4.5 Missing Values & Obvious Data-Quality Issues

Area	Observation	Week 1 Treatment / Implication
Customer dataset	No missing cells; no duplicate rows identified.	No immediate structural issue identified from Week 1 profiling.
Transaction — Device_Type	96 missing values.	Document for later treatment; do not assume a device without evidence.
Transaction — Location	96 missing values.	Document for later treatment; do not infer location at Week 1.
Identifiers	Customer_ID and Transaction_ID are unique in their respective datasets.	Use IDs for joins and counting; do not use Customer_Name as a unique identifier.
Risk_Review_Flag	Binary Yes/No label.	Treat as a synthetic educational label, not a confirmed fraud determination.
Amount_NGN	Positive values; range ₦100.17–₦693,454.45.	Later analysis should inspect the distribution and unusually large values.

Week 1 scope control

The assignment instructs interns not to perform extensive data cleaning yet. Therefore, this report records the observed quality issues and their implications; remediation will be handled during the data-preparation phase.

5. Track-Specific Analysis Planning

5.1 Analytical Questions

Area	Analytical Question	Potential Analysis
Customer behaviour	How does transaction activity vary across Customer_Segment groups?	Transaction count/value by segment.
Transaction activity	Which transaction types and channels are used most frequently?	Counts and shares by type and channel.
Transaction value	Which transaction types and channels contribute the highest total and average transaction value?	Total, average and distribution of Amount_NGN.
Customer value	Which customer segments contribute the highest total and average transaction value?	Join customer attributes to transaction measures.
Transaction status	What proportion of transactions are Successful, Failed, Pending or Reversed, and how does this vary by channel?	Status mix and status rate by channel.
Channel behaviour	How does transaction behaviour differ across Mobile App, USSD, Web, ATM and POS?	Volume/value/status comparisons across channels.
Risk patterns	What transaction characteristics are	Risk flag rate by transaction

	associated with transactions marked for risk review?	characteristics.
Customer-risk relationship	How do customer characteristics such as segment, account type, tenure and digital engagement relate to risk-review patterns?	Risk flag rate by customer attributes.

5.2 KPI Definition Table

KPI	Definition	Why It Matters	Required Data
Total Customers	Number of unique customers	Shows customer-base size and provides the denominator for customer analysis.	Customer_ID
Total Transactions	Number of recorded transactions	Measures overall transaction activity.	Transaction_ID
Total Transaction Value	Sum of Amount_NGN	Measures the total value processed during the analysis period.	Amount_NGN
Average Transaction Value	Total transaction value ÷ total transactions	Shows the average size of a recorded transaction.	Amount_NGN
Transaction Success Rate	Successful transactions ÷ total transactions × 100	Monitors transaction processing performance.	Transaction_Status
Risk Review Rate	Transactions flagged Yes ÷ total transactions × 100	Monitors the proportion of transactions carrying the synthetic review label.	Risk_Review_Flag
Average Transactions per Customer	Total transactions ÷ unique customers	Shows average transaction activity per customer.	Customer_ID, Transaction_ID
Average Digital Engagement Score	Average Digital_Engagement_Score across customers	Provides a high-level measure of customer digital engagement.	Digital_Engagement_Score

6. Dashboard Planning

The Week 1 requirement is a dashboard wireframe rather than a fully developed Power BI dashboard. The proposed design below prioritises management-level KPIs first, followed by customer, transaction and risk views.

TOTAL CUSTOMERS	TRANSACTIONS	TOTAL VALUE	SUCCESS RATE	RISK REVIEW RATE
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DATE	SEGMENT	ACCOUNT TYPE	CHANNEL	TXN TYPE	CITY / LOCATION
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CUSTOMER OVERVIEW Customer segment distribution Account type / status Digital engagement	TRANSACTION ACTIVITY Transaction volume over time Transaction type mix
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CHANNEL PERFORMANCE Volume and value by channel Status rate by channel	TRANSACTION STATUS Successful / Failed / Pending / Reversed Trend and channel comparison
TRANSACTION VALUE Total and average value Value by type and segment	RISK PATTERNS Risk review rate Patterns by type, channel and international status

6.1 Proposed Dashboard Filters

- Transaction Date / Month
- Customer Segment
- Account Type
- Customer City / Transaction Location
- Transaction Type
- Channel
- Transaction Status
- International Transaction

6.2 Expected Insights

The dashboard is expected to communicate evidence-based findings about customer activity, transaction volume and value, channel performance, transaction outcomes and patterns associated with the synthetic risk-review label. The actual findings will be determined after exploratory analysis rather than assumed in Week 1.

7. Initial Objectives & Success Criteria

7.1 Initial Objectives

7. Establish a reliable understanding of the customer and transaction datasets and their relationship.
8. Develop a focused set of analytical questions and KPIs that are answerable using the available data.
9. Build a Power BI-ready analytical structure that supports customer, transaction, channel and risk-pattern analysis.
10. Produce clear, evidence-based business insights that can support FinTrust stakeholders.

7.2 Success Criteria

- The two datasets are correctly related through Customer_ID.
- Data types, missing values, duplicates and obvious quality issues are documented and addressed appropriately during the preparation phase.
- The selected KPIs can be calculated consistently and reconciled against the underlying data.
- The dashboard communicates customer, transaction, channel and risk-related patterns clearly.
- Insights are supported by calculated evidence and are not based on unsupported assumptions.
- Risk-review findings are presented as patterns in a synthetic educational label, not as confirmed fraud determinations.

8. Risks & Future Dependencies

Risk / Dependency	Evidence	Potential Impact	Mitigation / Next Step
Missing Device_Type / Location	96 missing values in each field.	May affect	Profile impact; decide treatment

		channel/device/location analysis.	during Week 2.
Synthetic risk label	Risk_Review_Flag is educational/synthetic.	Could be misinterpreted as confirmed fraud.	Use careful language and document the limitation.
Data relationship errors	Customer_ID is critical for joining datasets.	Incorrect joins could distort customer-level analysis.	Validate key integrity before modelling/dashboarding.
Large transaction values	Amount_NGN has a wide range.	A small number of high-value transactions could influence averages.	Use distributions and complementary statistics.
Scope expansion	Additional questions may arise during exploration.	Could delay core deliverables.	Prioritise Week 1 questions and KPIs.

9. Week 2–4 Plan

Week	Planned Activities
Week 2 — Data Preparation & EDA	Validate and prepare the data; address documented quality issues; confirm joins; create useful date fields; perform exploratory analysis; establish baseline KPI calculations.
Week 3 — Analysis & Dashboard	Develop customer, transaction, channel and risk-pattern analyses; build the Power BI dashboard; validate measures and filters; refine visual design.
Week 4 — Insights & Finalisation	Translate analytical findings into business insights; validate results; document methodology and limitations; finalise dashboard and presentation materials.

10. Professional Documentation & Working Principles

- Use clear field names, KPI definitions and documented assumptions.
- Keep raw data separate from prepared/analytical data where possible.
- Record important data-quality decisions rather than silently changing records.
- Use reproducible calculations and reconcile dashboard KPIs to the source data.
- Distinguish observed facts from interpretation and recommendations.
- State limitations whenever a result depends on incomplete or synthetic information.
- Maintain a versioned project folder and update the project repository as required by the internship.

Interpretation principle

The purpose of Week 1 is to establish what the data can realistically answer. Findings and recommendations should be generated from the later analysis rather than predetermined.

Appendix A — Key Dataset Snapshot

Metric	Verified Week 1 Value
Customer records	1,500
Customer fields	12
Transaction records	12,000
Transaction fields	11

Unique customers	1,500
Unique transactions	12,000
Average transactions per customer	8.0
Transaction period	01 Jan 2026 – 31 Mar 2026
Total transaction value	₦560,477,354.85
Average transaction value	₦46,706.45
Successful transactions	10,856 (90.5%)
Risk review flag = Yes	2,352 (19.6%)
Missing Device_Type	96
Missing Location	96

Appendix B — Week 1 Submission Checklist

- ☐ Business understanding
- ☐ Resource review
- ☐ Track-specific analysis/planning
- ☐ Initial objectives
- ☐ Success criteria
- ☐ Week 2–4 plan
- ☐ Relevant risks
- ☐ Relevant future dependencies
- ☐ Professional documentation